

Coding & Solution

Date	8 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/pathuritejasri-netizen/smart-lender-ml
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Bootstrap 5
Key Features Implemented	Interactive Web Dashboard: Clean Bootstrap 5 form interface designed for users and credit officers to input application credentials easily.

	Real-Time Preprocessing Engine: Implements automated missing value imputation and robust feature scaling matching the dataset's variables. XGBoost Inference Server: Serves the top-performing, highly optimized XGBoost prediction pipeline backend directly via serialized models. Dynamic Status Alerts: Visual indicator system built directly into the UI, yielding instant applicant risk approvals or high-risk flags.
Pending / Incomplete Features	None. All target exploratory visualization benchmarks, preprocessing stages, model testing steps, and live Flask routing components are fully integrated and functional.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

intercepted gracefully without causing \$500\$ Internal Server errors.